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ATIS BROADCAST INFORMATION BRAVO // AVIONICS SCENARIO FLIGHT EXECUTION DIRECTIVE
STATION: VAPO SITL   |  FLEET: ARDUPILOT EKF3 FIXED-WING & VTOL AIRCRAFT
OPERATIONAL DIRECTIVE: GPS DENIAL, ELECTRONIC JAMMING, SPOOFING GATING & DEAD RECKONING
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PART I: Avionics Parameter Technical Reference

1. GNSS Denial, Hardware Cutoff & Jamming Simulation

SIM_GPS1_ENABLE **Type:** SITL Hardware Toggle
Default: 1 (Active) → **Execution Target:** 0 (Severed / Denied)
MCDU / Console Command: `param set SIM_GPS1_ENABLE 0`

Description: Simulates complete power cutoff or physical antenna severance. SITL immediately drops satellite tracking ($fix_type = 0$, $sats = 0$). EKF3 immediately declares horizontal GPS invalid and transitions directly to smooth Dead Reckoning without coordinate jitter.

SIM_GPS1_JAM **Type:** SITL Electronic Warfare Injection
Default: 0 (Normal RF) → **Execution Target:** 1 (RF Jammed)
MCDU / Console Command: `param set SIM_GPS1_JAM 1`

Description: Injects broadband RF interference on GPS L1/L2 carrier frequencies. Receiver C/N_0 drops below lock threshold (< 28 dB-Hz), inflating pseudorange noise. **Caution:** In SITL, the receiver retains a degraded 3D fix with jumping coordinates; requires automated gating configuration (Section 2) to prevent aircraft wandering.

2. Spoofing Defense, Innovation Gating & Attitude Protection

EK3_GLITCH_RAD **Type:** Innovation Step Gate
Default: 25.0 m → **Recommended Setting:** 15.0 to 20.0 m
MCDU / Console Command: `param set EK3_GLITCH_RAD 15`

Description: Defines maximum horizontal jump distance (meters) between consecutive updates before EKF3 rejects the measurement as anomalous. Rejects abrupt spoofed position jumps and extreme jamming noise spikes.

FS_EKF_THRESH **Type:** Flight Safety Failsafe Threshold
Default: 0.80 → **Anti-Jam Setting:** 0.60 to 0.65
MCDU / Console Command: `param set FS_EKF_THRESH 0.60`

Description: Normalized innovation variance threshold triggering the EKF Failsafe. Lowering this threshold to 0.60 forces ArduPlane to trip the failsafe immediately under jamming noise rather than tolerating noisy jumping fixes.

AHRS_GPS_USE **Type:** Attitude Filter Isolation
Default: 1 (Enabled) → **Hostile RF Target:** 0 (Isolate AHRS)
MCDU / Console Command: `param set AHRS_GPS_USE 0`

Description: Disconnects GPS ground velocity from attitude centrifugal turn compensation ($a_{\text{cent}} = \omega_z \cdot V_{\text{gps}}$). Prevents erratic GPS velocity spikes from corrupting the artificial horizon and inducing violent wing-rocking.

3. Dead Reckoning & Wind Estimation Configuration

EK3_DRAG_BCOEF_X / EK3_DRAG_BCOEF_Y **Type:** Aerodynamic Ballistic Coefficients
Default: 0.0 (Disabled) → **Airframe Target:** 25.0 to 35.0 (kg/m^2)
MCDU / Console Commands: `param set EK3_DRAG_BCOEF_X 25` — `param set EK3_DRAG_BCOEF_Y 25`

Description: Calibrates longitudinal and lateral airframe drag. In GPS-denied flight, EKF3 combines pitot true airspeed (V_{tas}) with accelerometer drag deceleration to estimate the horizontal wind vector ($\vec{W}$), establishing ground track: $\vec{V}_{\text{ground}} = \vec{V}_{\text{air}} + \vec{W}$.

`EK3_SRC_OPTIONS` **Type:** Sensor Fallback Bitmask
Default: `0` (Manual Fallback) → **Autonomous Target:** `1` (Auto Fallback)
MCDU / Console Command: `param set EK3_SRC_OPTIONS 1`

Description: Enables automatic sensor cascade. Setting Bit 0 allows EKF3 to drop an unhealthy GPS autonomously and switch to Dead Reckoning or Optical Flow without pilot intervention.

PART II: Tactical Flight Scenario Execution Protocols

SCENARIO 1: Total GPS Denial / Hardware Cutoff (Clean Dead Reckoning)

Operational Objective: Evaluate pure aerodynamic and inertial dead reckoning following complete loss of primary satellite reception (antenna severance or hardware failure). Verify that the aircraft maintains mission waypoints without wander.

Prerequisites & In-Flight Conditions:

- Airspeed sensor healthy and pitot tube calibrated (± 1.0 m/s).
- Aircraft in stable cruise in `AUTO` or `GUIDED` mode at 50–100 m AGL.
- Drag coefficients set: `EK3_DRAG_BCOEF_X = 25`, `EK3_DRAG_BCOEF_Y = 25`.

Step 1 — Pre-Test Verification: Verify groundspeed and airspeed telemetry match expected calm-air values ($V_{\text{tas}} \approx 18$ m/s).

Step 2 — Execution Command (MCDU Line 1L or SITL Console):

```
param set SIM_GPS1_ENABLE 0
```

Step 3 — Telemetry & Aircraft Flight Response:

- MAVLink `GPS_RAW_INT`: *fix_type* drops immediately to `0` (NO_FIX), *satellites_visible* drops to `0`.
- EKF3 detects instant satellite loss: horizontal GPS aiding is cleanly decoupled.
- Aircraft transitions into **Wind-Aided Dead Reckoning**: banks smoothly on flight plan course, holds ground track via pitot airspeed and compass heading. Zero weaving or wing-rocking occurs.

Step 4 — Recovery Sequence (Return to Satellite Navigation):

```
param set SIM_GPS1_ENABLE 1
```

Response: GPS lock reacquired ($fix_type = 3$); EKF3 fuses satellites back into navigation state within 3–5 seconds.

SCENARIO 2: Electronic Warfare RF Jamming Countermeasure (Anti-Hunt Protocol)

Operational Objective: Counter hostile broadband RF noise injection. Prevent the aircraft from chasing jumping coordinates (the "here-and-there" hunting phenomenon) by forcing automatic innovation rejection and prompt Dead Reckoning fallback.

Technical Root Cause of Wandering: During RF jamming, the receiver does *not* declare $fix_type = 0$ immediately. It outputs a degraded 3D fix with jumping coordinates. The L1 navigation controller tries to turn toward each false jump, and `AHRS_GPS_USE 1` tilts the horizon with false ground velocities.

Step 1 — Execute Anti-Jamming Defensive Pre-Configuration: Execute the following parameters in order via the MCDU or Console:

Command	Operational Function
`param set EK3_SRC_OPTIONS 1`	Enables EKF3 automatic fallback away from failing GPS.
`param set FS_EKF_THRESH 0.60`	Tightens failsafe threshold to trip instantly under RF noise.
`param set EK3_GLITCH_RAD 15`	Tightens innovation gate from 25 m to 15 m.
`param set AHRS_GPS_USE 0`	Severs GPS velocity from attitude filter to prevent wing-rocking.

Step 2 — Execution Command (Inject RF Jamming Noise):

```
param set SIM_GPS1_JAM 1
```

Step 3 — Telemetry & Aircraft Flight Response:

- SITL injects carrier noise: receiver HDOP inflates rapidly (> 4.5) and position variance surges.
- With `FS_EKF_THRESH 0.60`, the EKF failsafe trips within 4–6 seconds.
- With `EK3_SRC_OPTIONS 1`, EKF3 severs the noisy GPS and cascades into Dead Reckoning.
- With `AHRS_GPS_USE 0`, the wings remain completely level; erratic weaving is eliminated.

Step 4 — Recovery Sequence (Cease Jamming & Restore Navigation):

```
param set SIM_GPS1_JAM 0   →   param set AHRS_GPS_USE 1
```

SCENARIO 3: Adversarial GPS Spoofing / Innovation Step-Jump Defense

Operational Objective: Validate EKF3 innovation gating defense against hostile terrestrial spoofers attempting to hijack the aircraft trajectory via false pseudoranges or sudden coordinate offset steps.

Step 1 — Gating Defensive Pre-Configuration:

`param set EK3_GLITCH_RAD 15` — `param set AHRS_GPS_USE 0`

Step 2 — Execution Command (Inject Adversarial Coordinate Step):

`param set SIM_GPS1_GLT_X 60` — `param set SIM_GPS1_GLT_Y 60`

Alternative MCDU Action: Stage offset injection via Cockpit MCDU Line 3L.

Step 3 — Telemetry & Aircraft Flight Response:

- The spoofed coordinate jump (≈ 85 m) exceeds the `EK3_GLITCH_RAD` 15 m gate.
- EKF3 flags `GPS_Glitch: pos step` in MAVLink `STATUSTEXT`; the false position vector is discarded.
- The aircraft continues holding planned track on internal inertial dead reckoning, completely ignoring the spoofer's pull.

Step 4 — Recovery Sequence (Cease Spoofing Signal):

`param set SIM_GPS1_GLT_X 0` — `param set SIM_GPS1_GLT_Y 0` — `param set AHRS_GPS_USE 1`

SCENARIO 4: VTOL / Low-Altitude GPS Failure with Optical Flow Fallback

Operational Objective: Ensure stationary hover and station-keeping capability for VTOL platforms or low-altitude quadplanes when GPS is denied in zero-airspeed hover flight.

Step 1 — Sensor Pre-Configuration:

`param set SIM_FLOW_ENABLE 1` — `param set EK3_FLOW_USE 1` — `param set EK3_SRC1_VELXY 5`

Step 2 — Flight Regime Establishment: Establish the aircraft in steady hover (`QLOITER` or `QHOVER` mode) at 5–15 m AGL over textured ground.

Step 3 — Execution Command (Sever GPS in Hover):

`param set SIM_GPS1_ENABLE 0`

Step 4 — Telemetry & Aircraft Flight Response:

- MAVLink `OPTICAL_FLOW` packets stream angular ground pixel displacement rates (rad/s).
- EKF3 seamlessly substitutes satellite horizontal velocity with integrated optical flow ground velocity.
- Aircraft holds stationary hover coordinates without drift, even with zero forward airspeed over the pitot tube.

Step 5 — Recovery Sequence:

`param set SIM_GPS1_ENABLE 1` — `param set EK3_SRC1_VELXY 3`

SCENARIO 5: High-Crosswind Atmospheric Turbulence Dead Reckoning Validation

Operational Objective: Validate EKF3's wind triangle estimation ($\vec{V}_{\text{ground}} = \vec{V}_{\text{tas}} + \vec{W}_{\text{wind}}$) and crab-angle compensation under strong crosswind gusts and Dryden turbulence during GPS-denied navigation.

Step 1 — Inject Meteorological Crosswind Environment:

`param set SIM_WIND_SPD 12.0` — `param set SIM_WIND_DIR 270` — `param set SIM_WIND_TURB 0.25`

Step 2 — Execution Command (Sever GPS under Crosswind):

`param set SIM_GPS1_ENABLE 0`

Step 3 — Telemetry & Aircraft Flight Response:

- EKF3 estimates the 12 m/s westerly wind vector from frontal drag deceleration and sideslip drag dynamics.
- Navigation controller automatically establishes crab angle: $\beta = \arcsin\left(\frac{W_{\perp}}{V_{\text{tas}}}\right) \approx 35^{\circ}$ to 40°.
- Ground track remains aligned with planned waypoint course line despite turbulent atmospheric buffet.

Step 4 — Recovery Sequence:

`param set SIM_GPS1_ENABLE 1` — `param set SIM_WIND_SPD 0` — `param set SIM_WIND_TURB 0`

PART III: Master Flight Test Matrix & Telemetry Checksheet

Scenario	Pre-Requisite Setup	Execution Trigger	Key Telemetry Sign	Evaluation Criteria
1. Clean GPS Loss	Airspeed calibrated, `BCOEF_X/Y = 25`	`param set SIM_GPS1_ENABLE 0`	`GPS.fix = 0` `GPS.sats = 0`	Smooth transition to Dead Reckoning; zero wandering or wing-rock.
2. RF Jamming Counter	`EK3_SRC_OPTIONS 1` `FS_EKF_THRESH 0.60` `AHRS_GPS_USE 0`	`param set SIM_GPS1_JAM 1`	`HDOP > 4.5` `FS_EKF` triggered within 6 seconds	EKF failsafe trips promptly; drops degraded GPS; zero coordinate hunting.
3. Spoofing Defense	`EK3_GLITCH_RAD 15` `AHRS_GPS_USE 0`	`param set SIM_GPS1_GLT_X 60`	`GPS_Glitch` in STATUSTEXT	Measurement rejected at gate; internal dead reckoning maintained.
4. Optical Flow Hover	`SIM_FLOW_ENABLE 1` `EK3_SRC1_VELXY 5`	`param set SIM_GPS1_ENABLE 0` (in hover)	`OPTICAL_FLOW` packet rate $>$ 20 Hz	Zero positional drift in low-altitude hover; velocity aided by camera.
5. High-Wind DR	`SIM_WIND_SPD 12` `SIM_WIND_DIR 270`	`param set SIM_GPS1_ENABLE 0`	Crab angle established; ground track true	Correct crosswind crab computed from drag model; path held in turbulence.

Cockpit MCDU Quick-Key Map Reference

- **Line 1L (< SIM GPS OFF):** Stages `param set SIM_GPS1_ENABLE 0` for instant clean dead reckoning test.
- **Line 2L (< RF JAM 100%):** Stages `param set SIM_GPS1_JAM 1` for electronic warfare noise testing.
- **Line 3L (< LAT/LON SPOOF):** Stages coordinate step glitch to test innovation gating.
- **Line 4L (< WIND DR ON):** Stages `EK3_DRAG_BCOEF_X 25` and `EK3_DRAG_BCOEF_Y 25`.
- **Line 5L (< ANTI-JAM CONFIG):** Stages `EK3_SRC_OPTIONS 1`, `FS_EKF_THRESH 0.60`, and `AHRS_GPS_USE 0` to eliminate hunting.
- **Line 6R (< RECOVER ALL):** Instantly restores all parameters to nominal GPS-guided flight.